

Group Information

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L03: Electroencephalography I (EEG)

I. Data and Calculations

Subject Profile

Name: 蔣易翰

Height (cm):

Gender: Male

Age:

Weight (kg):

Dominant arm: Right

A. EEG Amplitude Measurements from Standard Deviation measurements (uV)

Table 3.1 Standard Deviation [STD]: for FP1-FP2-A1 set

Rhythm	CH Measurement	Eyes Closed	Eyes Open	Eyes Re-closed
Alpha	40 Stddev	1.72	2.60	2.72
Beta	41 Stddev	1.85	1.95	2.47
Delta	42 Stddev	4.96	12.30	9.31
Theta	43 Stddev	2.39	4.89	4.00

Table 3.2 Standard Deviation [STD]: for BIOPAC set

Rhythm	CH Measurement	Eyes Closed	Eyes Open	Eyes Re-closed
Alpha	40 Stddev	7.30	5.76	5.59
Beta	41 Stddev	6.94	5.92	5.26
Delta	42 Stddev	4.75	5.80	7.05
Theta	43 Stddev	6.56	4.89	5.82

B. EEG Frequency Measurements from first 'Eyes closed' data

Table 3.3 Frequency (Hz): for FP1-FP2-A1 set

Rhythm	CH Measurement	Cycle 1	Cycle 2	Cycle 3	Mean
Alpha	SC Freq	9.11	10.44	9.78	9.78
Beta	SC Freq	18.67	16.78	19.00	18.15
Delta	SC Freq	1.67	1.67	1.44	1.59
Theta	SC Freq	5.44	5.00	5.11	5.18

Table 3.4 Frequency (Hz): for BIOPAC set

Rhythm	CH Measurement	Cycle 1	Cycle 2	Cycle 3	Mean
Alpha	SC Freq	10.11	8.78	8.78	9.22
Beta	SC Freq	14.56	15.67	15.11	15.11

Delta	SC	Freq	2.78	2.33	2.72	2.72
Theta	SC	Freq	5.78	6.33	6.11	6.07

II. Questions

A. Compare and contrast synchrony and alpha block.

Table 3.5 Compare and contrast synchrony and alpha block

	Synchrony	Alpha block
By definition	Neuronal firing rhythms are consistent, with similar and regular alpha wave frequencies, and the amplitudes overlap, resulting in a stronger signal.	When the eyes are open, different neurons process different information. Alpha waves decrease and are replaced by low-amplitude, high-frequency beta waves.
In the FP1-FP2-A1 set experiment <i>Alpha wave STD:</i> $1.72 \mu V$ (eyes closed) < $2.60 \mu V$ (eyes open)	Data did not match the theory. Possible reasons: 1. Poor electrode contact (lack of pressure) led to signal attenuation. 2. The subject was not truly relaxed (remaining alert, calculating, or tense) when their eyes were closed.	Data did not match the theory. Possible reasons: 1. Subjects may be relaxed or have blurred vision when their eyes are open. 2. Blinking or eye movement caused noise interference. 3. The frontal lobe responds weakly to visual stimuli.
In the BIO PAC set experiment <i>Alpha wave STD:</i> $7.30 \mu V$ (eyes closed) >> $6.76 \mu V$ (eyes open)	Data met theoretical expectations. This electrode assembly is located close to the visual cortex and can clearly capture the enhancement of alpha waves when the eyes are closed.	Data are met with expectations. The visual cortex is activated upon receiving visual input, while alpha wave activity is suppressed.

B. Examine the alpha and beta waveforms for change between the “eyes closed” state and the “eyes open” state based on BIOPAC tri-electrode set.

i. Does **desynchronization** of the alpha rhythm occur when the eyes are open?

YES. The data in Table 3.2 show that:

- Eyes Closed: α wave STD = $7.30 \mu V$
- Eyes Open: α wave STD = $6.76 \mu V$ (decreased by $1.85 \mu V$)

ii. Does the beta rhythm become more pronounced in the “eyes open” state?

NO. The data in Table 3.2 show that:

- Eyes Closed: β wave STD = $6.94 \mu V$
- Eyes Open: β wave STD = $5.92 \mu V$ (decreased by $1.77 \mu V$)

Theoretically, when a subject opens their eyes and their brain enters an alert state, the amplitude of beta waves should increase.

Possible reasons for the discrepancy between the experiment and the theory include:

The subject was not truly relaxed when they first closed their eyes and was still thinking, or the EMG generated by swallowing saliva overlapped with the beta frequency band.

- C. The amplitude measurements (STD) are indicative of how much alpha activity is occurring in Subject. But, the amplitude values for beta do not truly reflect the amount of mental activity occurring with the eyes open. Explain.

Table 3.6 Compare the effects of alpha and beta wave amplitudes (STD) on psychological activity

Alpha wave	Bate wave
- The alpha wave band (8-13 Hz) is relatively narrow and stable. - Corresponding to the relaxed state, the causal relationship is clear. - When the eyes are closed, alpha waves dominate the brain signals, and brain activity is relatively simple. Therefore, alpha waves CAN indicate whether a subject is relaxed.	- The beta wave band (13-30 Hz) is wide and cluttered. - Interference containing non-neural signals, such as EOG, EMG, and ECG interference. - When the eyes are open, the brain desynchronizes and neurons become active and process different information individually, resulting in different firing phases that cancel each other out. Therefore, beta waves CANNOT truly reflect the amount of mental activity.

To accurately assess the intensity of a subject's mental activity when their eyes are open, it is recommended to train the subject to perform specific cognitive tasks and conduct time-frequency analysis using **continuous wavelet transform** and **fast Fourier transform**.

Fig 1. Biopac set Spectrogram

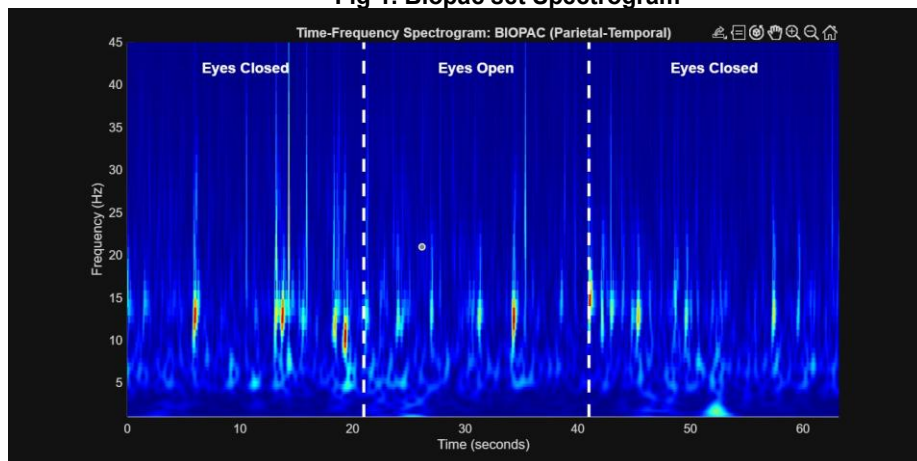
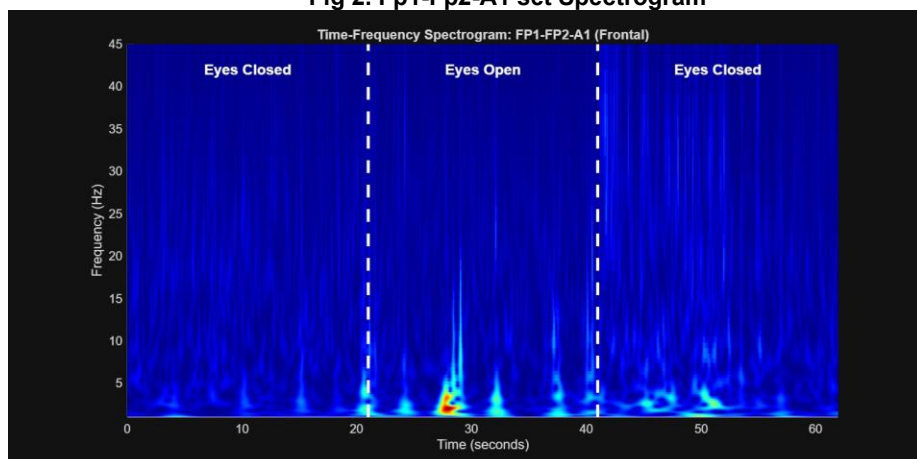


Fig 2. Fp1-Fp2-A1 set Spectrogram



- D. Examine the delta and theta rhythm. Is there an increase in delta and theta activity when the eyes are open? Explain your observation [based on BIOPAC tri-electrode set](#).

Theoretically, delta waves primarily occur during deep sleep, while theta waves are associated with relaxation or light sleep. These two low-frequency brainwave activities should not increase when the patient is awake and eyes are open.

Table 3.7 Compare the changes in amplitude (STD) of delta and theta waves across three periods: [for BIOPAC set](#)

	Delta waves	Theta waves
Eyes closed - Eyes open	Increased (from 4.75 μV to 5.80 μV) It is speculated that the electrodes are placed near the temporal lobe and neck, and the low-frequency noise generated by swallowing saliva or blinking and eye movements causes changes in electrode contact, which are mistaken for delta waves.	Decreased (from 6.56 μV to 4.89 μV) This is physiologically expected, as mild theta activity may occur when the eyes are closed and relaxed, and increased attention after opening the eyes will suppress theta waves, which is consistent with normal physiological responses.
Eyes open - Eyes Re-closed	Increased (from 5.80 μV to 7.05 μV) When the Delta wave closes its eyes again, the muscles are pulled again, causing low-frequency artifacts.	Increased (from 4.89 μV to 5.82 μV) A slightly higher value may reflect further relaxation or decreased attention in the subject, which is consistent with the physiological characteristics of a relaxed state.

Therefore, we believe that delta is subject to greater interference in the measurement and may not reflect true changes in neural activity, while the changes in theta are more consistent with normal brainwave responses.

- E. Compare the EEG amplitude and frequency characteristics in the [FP1-FP2-A1 set](#) vs. [BIOPAC set](#) across the three conditions (eye-closed, eye-open, eye-closed again).
- Which electrode set** shows a more pronounced change in alpha wave activity (8–13 Hz) when transitioning from eye-open to eye-closed? Why might this be the case?

BIOPAC set. When the eyes are closed, alpha increases significantly (+1.54), consistent with typical alpha synchronization.

Table 3.8 Compare Alpha wave amplitude (STD)

Electrode set	Eyes open (μV)	Eyes closed (μV)	Change (μV)
FP1-FP2-A1 set	2.60	1.72	-0.88
BIOPAC set	5.76	7.30	+1.54

On the contrary, in the FP1-FP2-A1 set, alpha decreased by -0.88 μV when the eyes were closed, which was completely contrary to the theory.

This may be because the BIOPAC set is located closer to the visual cortex (occipital/parietal lobe) and is more sensitive to changes in visual input; while the FP1-FP2-A1 set is located in the frontal lobe and is mainly related to cognitive function, showing a weaker response to simple eye opening/closing.

- ii. Considering the brain regions covered by each tri-electrode set, how do your observations relate to known neurophysiological functions of the frontal vs parietal-temporal lobes during rest and visual attention?

BIOPAC set (cover parietal-temporal lobe)

Neurophysiological function: The parietal and occipital lobes contain the primary and secondary visual cortex, processing visual information in the brain. Alpha waves in this area are considered the idling rhythm of the visual cortex.

Observation:

Eyes closed (rest): The visual cortex receives no visual input and is in a relaxed but synchronized state. Numerous neurons fire at a consistent rhythm, resulting in high-amplitude alpha waves.

- This corresponds to the idle state of the sensory cortex during rest.

Eyes open (visual attention): Visual stimuli enter, activating neurons in the visual cortex, which begin processing different visual features. This asynchronous activity disrupts the consistent alpha rhythm, leading to a decrease in amplitude (desynchronization).

- This reflects the active state of the sensory cortex during visual attention.

FP1-FP2-A1 set (cover frontal lobe)

Neurophysiological function: The frontal lobe (especially the prefrontal lobe) is responsible for higher-order cognitive functions such as working memory, attentional control, decision-making, and inhibitory control. Unlike the sensory cortex, which directly processes primitive sensory input, it regulates and monitors information processing.

Observation:

Compared to the visual cortex, which has a single primary function (processing vision), the frontal lobe contains dozens of areas with different functions (such as Broca's area, the frontal oculomotor area, and the prefrontal cortex). When performing a task, these areas may not all fire synchronously; some are active while others are inhibited. This heterogeneity leads to the cancellation or attenuation of the overall electric field, resulting in a smaller amplitude.

- This reflects that the frontal lobe does not primarily process visual signals.

- iii. Why might the BIOPAC set show different EEG dynamics from the FP1-FP2-A1 set during the eye-open condition? What functions are these areas responsible for?

As previously discussed, the parietal and temporal lobes (BIOPAC set) handle multisensory and spatial integration. Opening the eyes triggers neuronal desynchronization in these regions, suppressing the prominent resting alpha rhythm and reflecting a shift toward active sensory processing. Conversely, the frontal lobes (FP1-FP2-A1 set) governs higher-order cognitive functions. Because its role is modulatory and task-dependent, its baseline neural firing is inherently less synchronized. Consequently, the mere act of opening the eyes produces minimal EEG amplitude changes in the frontal region.

Furthermore, the standard deviation (STD) of the BIOPAC set is consistently higher than that of the frontal set across both visual conditions, reflecting intrinsic regional signal differences:

Parietal-Temporal: Driven by strong dominant resting rhythms (e.g., alpha) that cause large amplitude fluctuations. This high signal variability persists even during the transition to desynchronization upon eye-opening.

Frontal: Lacks dominant resting rhythms. Its low-amplitude, heterogeneous baseline activity results in less fluctuation and a consistently lower STD, regardless of visual input.

- iv. Based on your observations, **which tri-electrode set would you recommend** for studying alpha wave modulation in visual attention tasks? Justify your answer with your findings.

BIOPAC set.

1. Table 3.2 shows that the BIOPAC tri-electrode set clearly distinguishes between alpha decrease (desynchronization) during eye opening (alertness) and alpha increase (synchronization) during eye closing (relaxation), consistent with the expected physiological response. Also, the BIOPAC set has relatively high data values (5–7 μV), making it less susceptible to noise interference.
2. The BIOPAC tri-electrode set is located close to the visual cortex (parietal/occipital lobe) and directly reflects alpha activity related to visual processing. However, the FP1-FP2-A1 set does not respond clearly to changes in visual state (alpha actually decreases when eyes are closed), because the frontal lobe is primarily involved in cognitive control rather than passive visual processing, making it unsuitable for simple visual attention tasks.

Additional Questions

1. The simple bipolar configuration was used in measuring EEG of a specific brain region in this experiment. Please compare and discuss the different kinds of configuration used in measuring EEG.

Table 3.9 Comparison of Common EEG Electrode Configurations

Configuration	Mechanism	Advantages	Disadvantages
Bipolar	Measures potential difference between two active electrodes (e.g., FP1 vs. FP2).	• Excellent common-mode noise rejection. • Highly effective for localizing focal brain activity.	• Cannot measure absolute amplitude. • Signal cancellation (shows zero) if both electrodes pick up the same wave.
Monopolar	Measures difference between one active electrode and a "neutral" reference (e.g., earlobe A1).	• Captures true amplitude and wave morphology clearly. • No signal cancellation from neighboring sites.	• No true "zero-voltage" body part exists. • A noisy reference electrode will contaminate all connected active electrodes.
Average Reference	Measures one active electrode against the mathematical average of all scalp electrodes.	• Provides a highly stable and unbiased reference point. • Minimizes the impact of a single noisy reference.	• Requires high-density EEG (64+ channels). • Average can be skewed by large local signals if too few channels are used.

2. It is a consensus that EEG has better temporal resolution than the functional MRI, however, functional MRI has better spatial resolution. Please explain.

The consensus regarding the resolutions of EEG and fMRI stems from the fundamentally different physiological signals each modality measures:

- **Temporal Resolution:**

EEG directly records the electrical activity of neurons. Since electrical signaling is almost instantaneous, EEG captures real-time dynamic neural responses with millisecond (ms) precision. In contrast, fMRI measures the BOLD (Blood Oxygen Level-Dependent) signal. Because blood flow takes several seconds to increase after neurons fire, fMRI is inherently slow.

- **Spatial Resolution:**

fMRI uses powerful magnetic fields to map the 3D structure and blood oxygenation of the brain, pinpointing active regions with millimeter (mm) precision. In contrast, EEG signals must travel through the cerebrospinal fluid, skull, and scalp. The skull diffuses and "smears" the electrical signals, making it extremely difficult to pinpoint the exact deep-brain source, limiting EEG's spatial resolution to centimeters.

References